Design the POST Endpoint for Creating a New Product

Understand how to design the POST endpoint to add new products to a REST API. Explore the request and response structures, including headers and body content, to ensure correct server processing. Gain practical insights into handling status codes, content types, and URI locations for newly created resources.

To add a new product to the collection represented by the URI, we can use the POST method.

Creating a new product

According to the specifications, “The POST method is used to request that the origin server accept the entity enclosed in the request as a new subordinate of the resource identified by the Request-URI in the Request-Line.”

The above definition means that:

- The request will include an entity. This is an important point for validation and validation-related response.
- The server must accept the enclosed or included entity as an item that can be added to the collection of the resource identified by the URI. This is important because it means the server can’t process it as a part of a different resource.

Ask

Based on the above points, the following question comes to mind: How does the server know the format/MIME type of the entity? The answer to this lies in the

structure of the POST request, i.e., the header and body.

Request/response structure

Based on the above points, let's look at the structure of the POST request and response.

Request header

The POST request header contains the following:

- **Method:** This is the HTTP method to be executed by the server. It'll be POST in this case.
- **URL/URI:** This is the identifying URI of the resource.
- **Version:** This is the version of the HTTP, i.e., 1.0, 1.1, and so on.
- **Content-Type:** This is the MIME type/format of the content.
- **Content-Length:** This is the length/size of the enclosed entity or content.
- **Accept:** This is the MIME type/format of the body of the response that the client will accept.

Request body

The body of the request contains the data to be processed in the format specified in the header. If the format of the data in the body doesn't match the format specified in the header, the server can reject the request.

Similarly, the POST response sent by the server will also have a header and a body. The POST response tells the client whether the request has been successful or not and, optionally, the entity that was created or the reason for the failure of the request.

Ask

The server has to keep the following in mind while sending the POST response:

- The server should return the entity as part of the response body if the entity has been created as a result of the processing of the request.
- The header should contain the URI specifying the location of the entity. Optionally, the body can also contain the location details.
- The response isn't cacheable by default unless the cache controls are present.
- The body is optional In case of response.

Response header

The most commonly used headers for a POST response are:

- **Cache-Control:** This defines parameters such as interval, and URL of caching server.
- **Content-Type:** This is the format or MIME type of content in the body. This should be the same as that of the `Accept` header of the request.
- **Content-Length:** This is the size of the content within the body.
- **Status:** This is the status of the request. If the request is processed successfully, the status will be `201 CREATED`.
- **Location:** This is the URI of the newly created entity. Location header is mandatory if the status is `201 CREATED`.

> **Note:** Among the above, `Status` and `Location` are most important as they indicate the successful processing of the request.

Response body

The body part is optional for POST. The server can return the newly created entity in the body. However, please keep in mind that some servers just send back the location as a part of the body. If the body is present, its MIME type should be the same as the MIME type specified by the Accept header of the request.

Ask

Request/response structure for POST API to add new product

Products API

Keeping the above points in mind, the request header for adding a new product will be as follows:

index.js Javascript (babel-node)

```
POST /api/products HTTP/1.1
Host: localhost:5000
User-Agent: curl/7.71.1
Content-Type: application/json
Content-Length: 20
Accept: application/json
```

Ask

The `User-Agent` header will be optional.

The request body sent by the client will be:

index.js Javascript (babel-node)

```
{
        "name": "rest-container",
        "description": "container for rest",
        "version": "2.2"
}
```

The response header provided by the server will be:

index.js Javascript (babel-node)

```
HTTP/1.1 201 CREATED
Content-Type: application/json
Content-Length: 92
Location: http://localhost:5000/api/products/3
```

Ask

The response body sent by the server will be:

index.js Javascript (babel-node)

```
{
     "location":"http://localhost:5000/api/products/3",
     "product":{
                 "id":"3",
                 "name": "rest-container",
                 "description": "container for rest",
                 "version": "2.2"
               }
}
```

Note: The data given above is just for the sake of example.

Ask